



MRS6130-P1812

mmWave Radar Module

User Guide

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1 Overview

1.1 Module introduction

MRS6130 is a module developed based on the new generation of ultra-low power single-chip 60 GHz 1T3R FMCW AiP radar sensor chip RS6130 from Possumic. The module integrates a 1T3R AiP radar chip and flash, enabling rapid application development for developers after direct power supply to the module. The module includes four low-power management modes: shutdown, standby, idle, and working. It not only features low power consumption in standby mode but also maintains low power consumption during sensing through our fast burst engine and high-performance RF transceiver. Its configuration and data acquisition are implemented via the UART interface. The module is divided into three subsystems:

- **mmWave RF and BB Subsystem:** This module encompasses all high-performance radar RF/analog and base band digital signal circuits. It also incorporates a linear frequency modulated pulse generator and a Tx-to-Rx state machine.
- **Processing and Application Subsystem:** It comprises a 32-bit RISC CPU and a radar hardware accelerator (HWA) for diverse application processing and task management. Our proprietary HWA IP is utilized for handling specific radar computations (such as FFT, CFAR, etc.), compression, as well as moving target detection and localization.
- **Power and Clock Subsystem:** RS6130 is a complex single-chip SoC with multiple functional systems. We utilize the on-chip integrated PMU system and clock subsystem for centralized management.

1.2 Module specifications

Table 1-1 Module Specifications

Specifications	Minimum	Typical	Maximum	Unit	Remarks
Radar Specifications					
Operating Frequency Band	58.0	-	64.0	GHz	
Continuous Frequency Scanning Bandwidth	-	6	-	GHz	
Number of Transmitting and Receiving Channels	1T3R			-	It can achieve different mode configurations such as 1T1R, 1T2R, and 1T3R
Equivalent Isotropic Radiation Power (EIRP)	-	17	-	dBm	
Horizontal FOV	-	120	-	Degree	
Vertical FOV	-	120	-	Degree	
Farthest Detection Distance	-	-	20	m	It depends on the RCS size of the detection target
Recent Detection Distance	-	-	0.01	m	Take human targets as a reference
Hardware Specifications					
Module Dimensions	-	18x12	-	mm ²	
VCC Supply Voltage	1.7	3.3	5.5	V	

VDDIO Supply Voltage	1.7	3.3	3.6	V	
Operating Temperature	-40	-	105	°C	Ambient temperature

1.3 Application scenarios

The MRS6130 module can be widely applied in typical AIOT intelligent perception scenarios represented by smart home appliances, intelligent security, smart health care, and intelligent cockpits, achieving high-precision and high-sensitivity detection of human body and object movements, micro-motions, breathing, and heartbeats:

- Human movement/presence detection
- Human body tracking and counting
- Human respiratory and heartbeat detection
- Gesture recognition
- Video doorbell
- Network Cameras
- Air conditioner
- Refrigerator
- Television

2 System Application Block Diagram

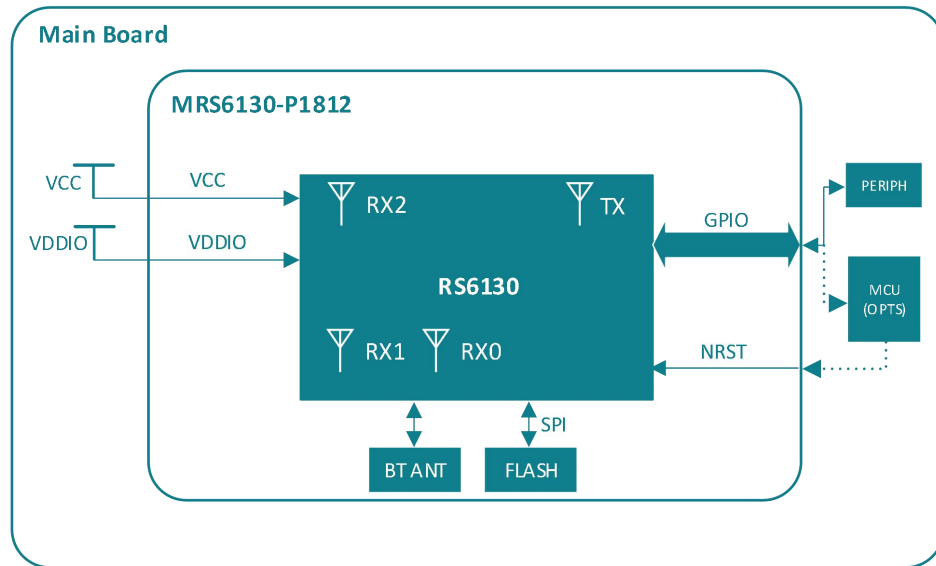


Figure 2-1 System Application Block Diagram

The MRS6130 millimeter-wave radar module's peripherals include a power supply and 10 GPIO. VCC is the main power supply for the module, while VDDIO is the power supply for the GPIO. When the power supply voltage does not exceed 3.6V, VCC and VDDIO can be directly short-circuited during application. V13 is powered by the internal VINT by default.

3 Pin Definition

3.1 Pin layout

The layout of the module's PIN is shown in the figure below.

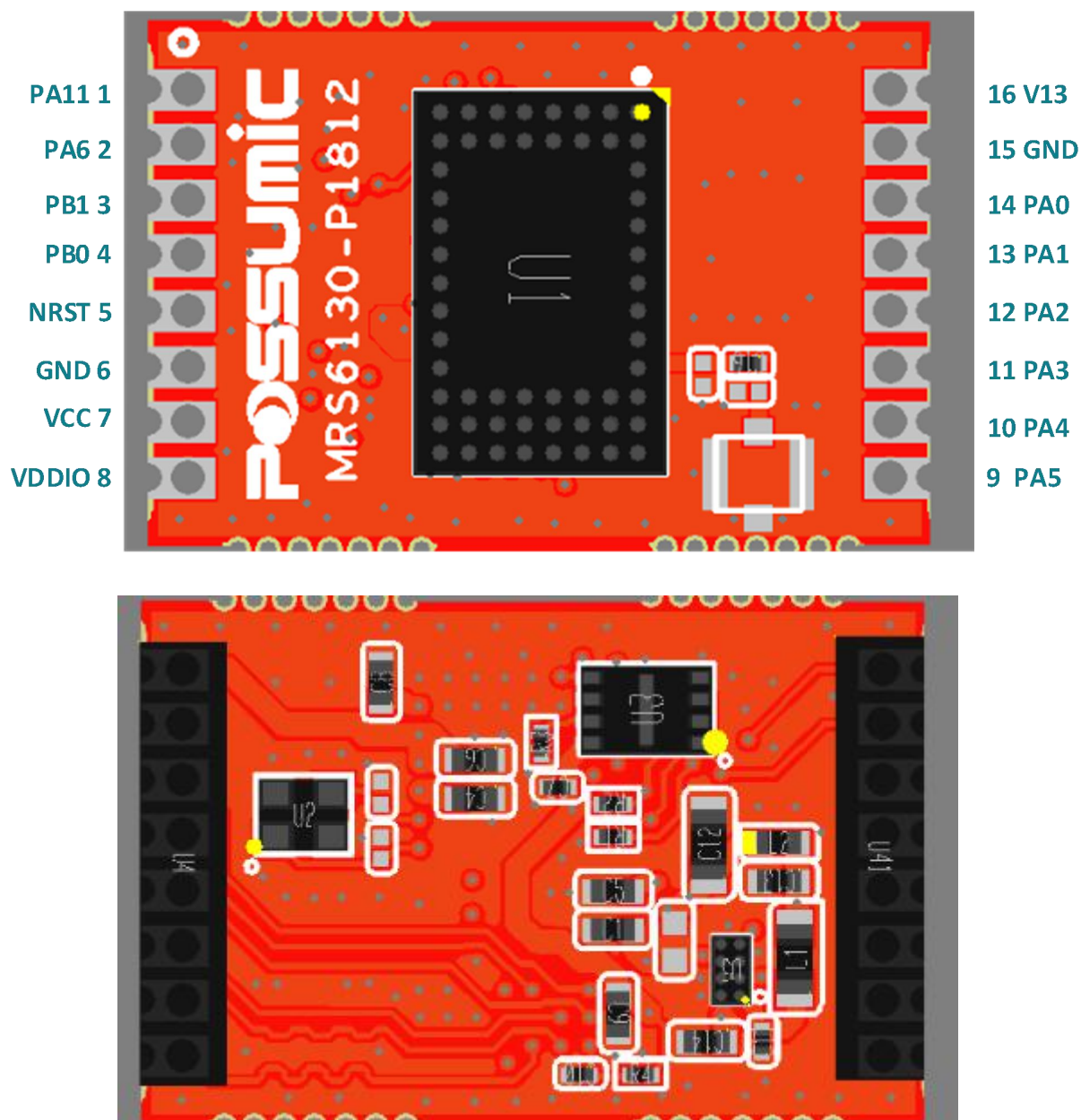


Figure 3-1 Module Pinout

3.2 Pin description

The module has 16 pins, and the functional descriptions of the pins are as shown in Table 3-1:

Table 3-1 Description of Module Pin Functions

Pin Name	Serial Number	Type	Functional Description
PA11	1	GPIO/WUP	WIO11,I2C1_SCL,PWM3
PA6	2	GPIO/WUP	WIO6,UART1_RTS,UART2_TX,I2C1_SDA,PWM6
PB1	3	GPIO/WUP	WIO17,UART0-RX,TWIO-SCL
PB0	4	GPIO/WUP	WIO16,UART0-TX,TWIO-SDA
NRST	5	I	Module reset
GND	6	G	Ground Pin, Short directly to board ground plane.
VCC	7	PI	The module operates with a power supply ranging from 1.7V to 5.5V, with a default input voltage of 3.3V
VDDIO	8	PI	The module IO power supply ranges from 1.7V to 3.6V, with a default input of 3.3V
PA5	9	GPIO/WUP	WIO5,SPI0_WP,I2C0_SCL,PWM5
PA4	10	GPIO/WUP	WIO4,SPI0_HOLD,I2C0_SDA,PWM4
PA3	11	GPIO/WUP	WIO3,SPI0_MISO,UART1_CTS,I2C0_SCL,UART2_RX,PWM3
PA2	12	GPIO/WUP	WIO2,SPI0_MOSI,UART1_RTS,I2C0_SDA,UART2_TX,PWM2
PA1	13	GPIO/WUP	WIO1,SPI0_CLK,UART1_RX,PWM1
PA0	14	GPIO/WUP	WIO0,UART1-TX,PWM0,GPADC7
GND	15	G	Ground Pin, Short directly to board ground plane.
V13	16	PI	DNU, maintain NC

Note:

- **G** Stand for the ground
- **PI** Stands for input power
- **PO** Stands for output power
- **GPIO** Stands for General-Purpose Digital Input/Output
- **I** Represents the input pin
- **WUP** Represents the digital pin that can maintain the state or wake up the system in standby mode, where **WIOx** Refers to the sequential number of the WUP type IO.

4 Peripheral

4.1 GPIO

The MRS6130-P1812 module provides 10 general-purpose GPIO, each GPIO pin features multiple switchable alternate functions. GPIO configuration features:

- Three modes: input, output, and high impedance
- Input configurable pull-up and pull-down
- The output drive capability is set to 20mA by default, with one setting at 20mA and a maximum configuration of 80mA
- IO output supports both push-pull mode and open-drain mode

Note:

- In low-power scenarios, all GPIO support triggering the system wake-up through interrupt mode.

4.2 UART

The MRS6130-P1812 module supports UART0 and UART1. UART is compatible with the 16550 industry standard and supports baud rates up to 3 Mbps. UART features include:

- Programmable transmit and receive baud rate
- Baud rate self-checking function
- Supports 64-byte receive and transmit FIFO,
- Supports data lengths of 5/6/7/8 bits
- Support 1/1.5/2 stop bits
- Support parity check
- Supports DMA high-speed data communication
- Supports UART wake-up function, with UART RX having the ability to detect rising edge changes and trigger interrupts. The number of rising edges detected can be configured from 1 to 8. In low-power scenarios, the system is woken up after UART RX receives a matching rising edge.

4.3 I2C

The MRS6130-P1812 module supports I2C interface communication, allowing users to choose between UART and I2C based on their specific requirements. The I2C interface supports the following features:

- Can be configured as master and slave modes
- Supports standard mode (100Kbit/s)
- Supports high-speed mode (400Kbit/s)
- Supports high-speed mode enhancement (1000Kbit/s)
- Supports 7-bit addressing mode and 10-bit addressing mode
- Supports receive and transmit FIFO depths of 8 bytes
- Supports I2C wake-up function. In low-power scenarios, I2C supports 7-bit addressing matching to trigger the wake-up system.

4.4 PWM

The PWM controller can be used to output modulated digital waveform signals, supporting an adjustable duty cycle from 0 to 100%.

4.5 SPI

RS6130 supports a quad-serial SPI interface (QSPI). The QSPI protocol supports half-duplex, full-duplex, synchronous, and serial communication with peripherals. This interface can be configured in either master or slave mode. There are four possible combinations of serial clock phase and polarity. The following features are supported:

- The clock frequency can reach up to 96 MHz
- Support Motorola SPI mode
- Programmable master mode or slave mode
- Supports standard SPI, dual SPI, and quad SPI modes
- Programmable data width, ranging from 4 to 32 bits
- The FIFO depth for both sending and receiving data is 32 words
- Programmable sampling delay time in main mode
- Programmable Serial Clock Polarity (SCPOL) and Serial Clock Phase (SCPH)
- Supports interrupt and DMA
- Programmable instructions, address length, and wait cycles in main mode
- Support clock detection to wake up the system from standby mode

5 Hardware Usage

5.1 Hardware start up

Module VCC is powered at 3.3V, VDDIO is powered at 3.3V, and V13 is internally powered.

NRST is first pulled low, then released to pull high, and the hardware is powered on to boot. The firmware is loaded from external Flash into SRAM for execution.

5.2 Firmware burning

Based on the specific hardware configuration, users can enter the burning mode of the chip through the following two schemes.

Option 1: The master can control the NRST pin through IO

- 1) Keep VCC powered on, with PB0 and PB2 at high voltage level;
- 2) Set NRST to low level and delay for 20ms;
- 3) Set NRST to high level and delay for 50ms;
- 4) Set PB0 (or PB2) to low level, and set NRST to low level;
- 5) Delay for 20ms, set NRST to high level;
- 6) Delay for 50ms, set PB0 (or PB2) to high level. At this point, the chip and flash have completed resetting and entered Burning mode.

Option 2: Suitable for systems with external control power supply interruption mechanisms

- 1) Power off our chips, Flash, or modules. If the external capacitor is 10uF, it is recommended to maintain the power-off state for 10 seconds. Please adjust the power-off duration based on the external capacitor and the actual power-down speed;
- 2) Set PB0 (or PB2) to low level;

Power on our chip and Flash or module again, delay for 50ms, and set PB0 (or PB2) to high level. At this point, the chip and flash have completed resetting and entered burning mode.

Finally, use BurnTool to program the firmware into the Flash on the module.

5.3 Demo interface

The U4 and U41 interfaces support docking with the MRS6130-AIP-DEV-V1 Demo development board.

6 Electrical Characteristics

6.1 Power supply operating conditions

Table 6-1 Operating Conditions of Module Power Supply

Name	Parameter Description	Minimum	Typical Value	Maximum	Unit
VCC	Module power supply input voltage	1.7	3.3	5.5	V
VDDIO	Module IO input voltage	1.7	3.3	3.6	V
V13	Internal power supply output voltage of the module	1.2	1.5	2.5	V
Ivcc	The current provided by the external power supply	-	-	500	mA

7 Module Size and PCB Packaging

7.1 Module dimensions

Unit:mm

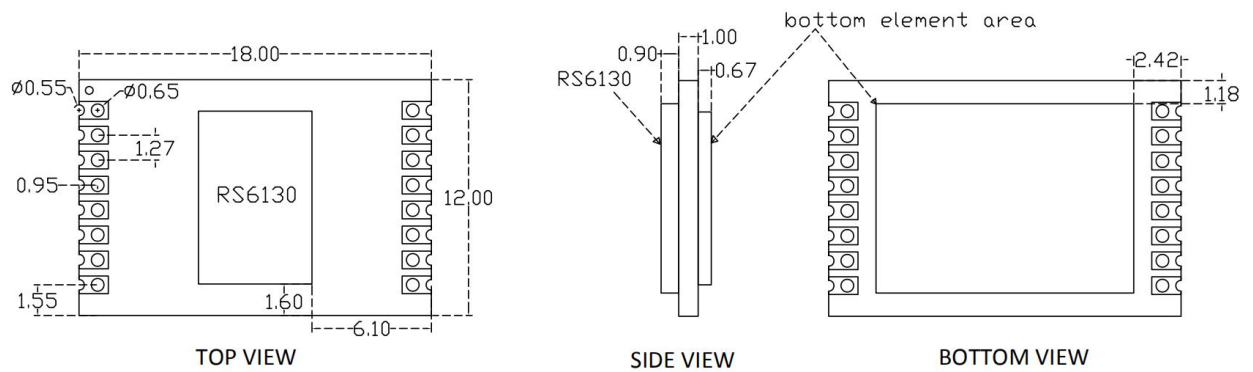


Figure 7-1 Dimensions of MRS6130-P1812 Module

7.2 Recommended PCB packaging

Unit:mm

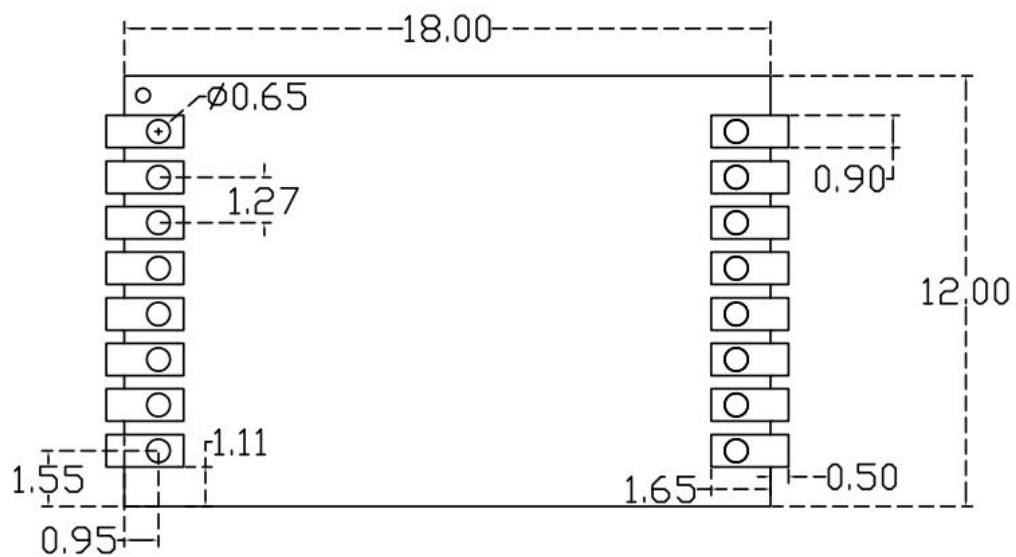


Figure 7-2 Recommended PCB Packaging

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